PostgreSQL Administration

1. Roles and access

Use separate application runtime and schema migration roles. Grant minimum privileges and rotate credentials through a secrets manager.

Practical guidance

Application role handles data access; migration role handles approved DDL.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

PostgreSQL Administration

2. Vacuum and analyze

Autovacuum reclaims dead tuple space for reuse and protects transaction ID health.

ANALYZE updates planner statistics.

Practical guidance

Investigate bloat and long-running transactions before tuning autovacuum thresholds.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

PostgreSQL Administration

3. Extensions and vectors

pgvector provides vector similarity functionality. Enable extensions in the target database only after compatibility and permissions checks.

Practical guidance

Example extensions: vector and pg_trgm; verify deployment requirements for each environment.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

PostgreSQL Administration

4. Backup and recovery

Combine managed snapshots or base backups with WAL archival where point-in-time recovery is required. Regularly validate restore procedures.

Practical guidance

Sample PostgreSQL RPO: 30 minutes. Sample RTO: 3 hours.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

PostgreSQL Administration

5. Monitoring and query plans

Use pg_stat_activity, pg_stat_statements when enabled, EXPLAIN (ANALYZE, BUFFERS) with caution, and storage metrics.

Practical guidance

Investigate locks, sequential scans, connection pressure, and replication lag in context.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.